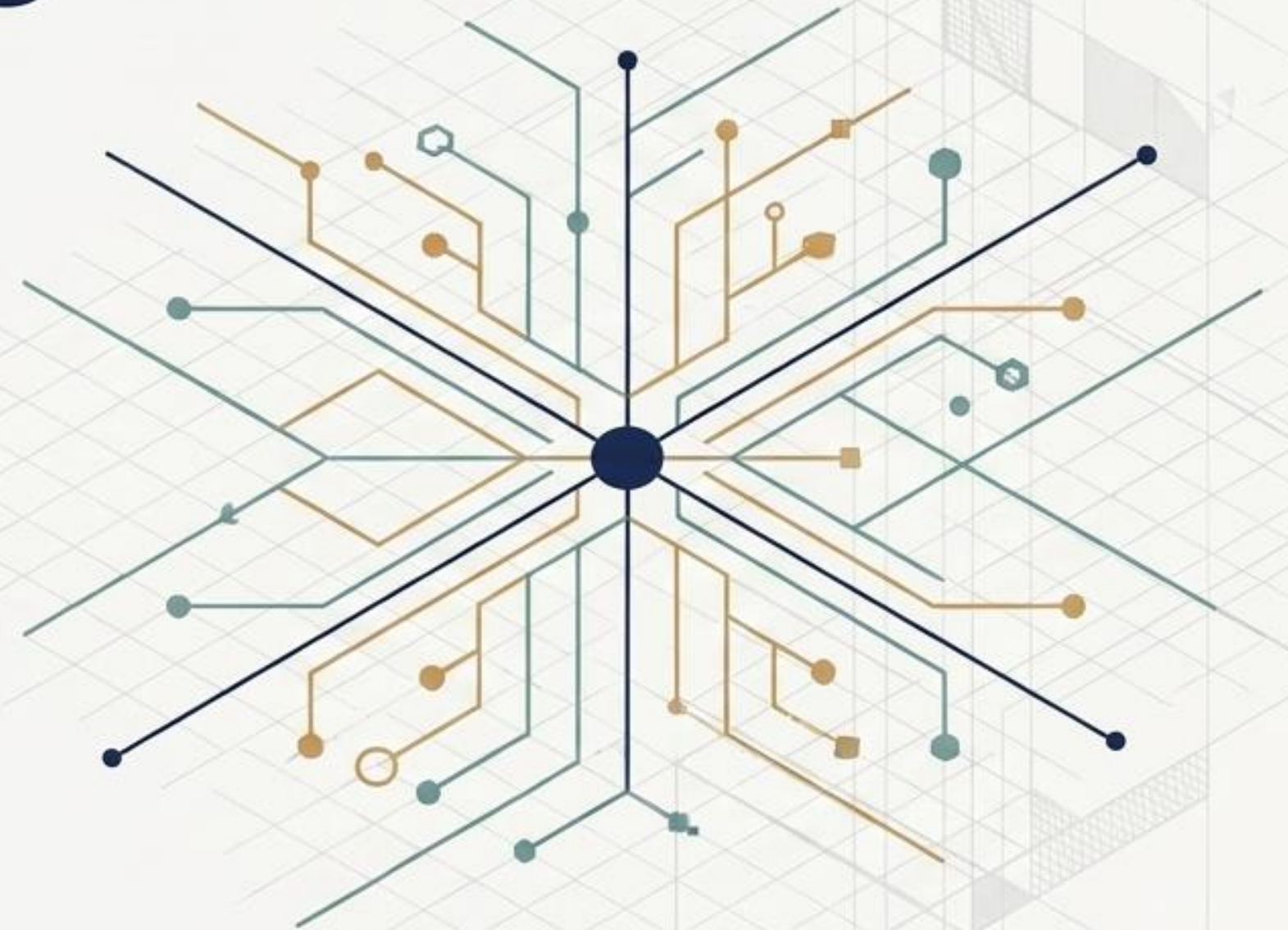


Understanding Language, Not Just Words

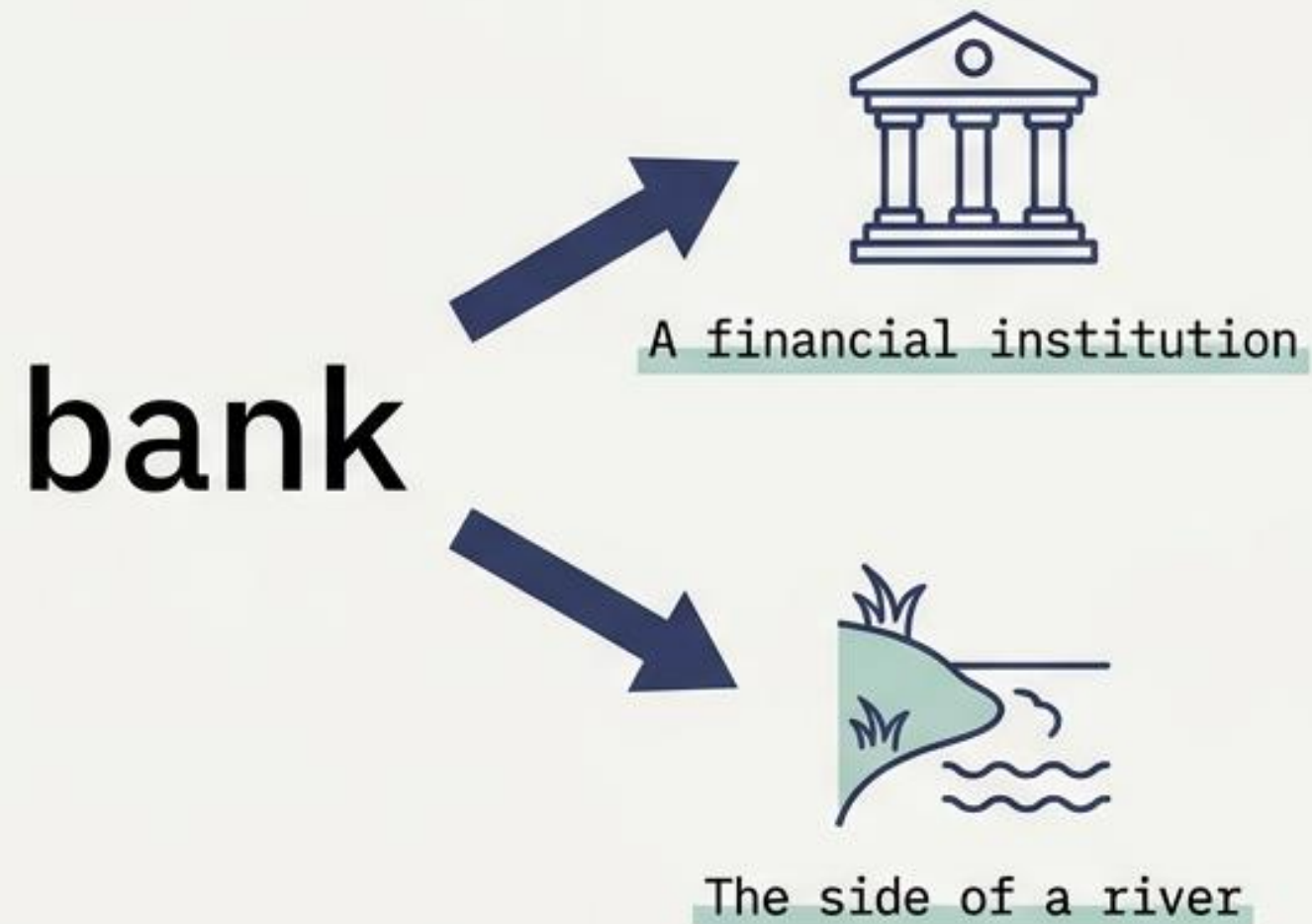
Demystifying BERT and the
Transformer Architecture

Based on research by Dr Partha Majumdar (ORC-ID: 0009-0002-7375-8034)

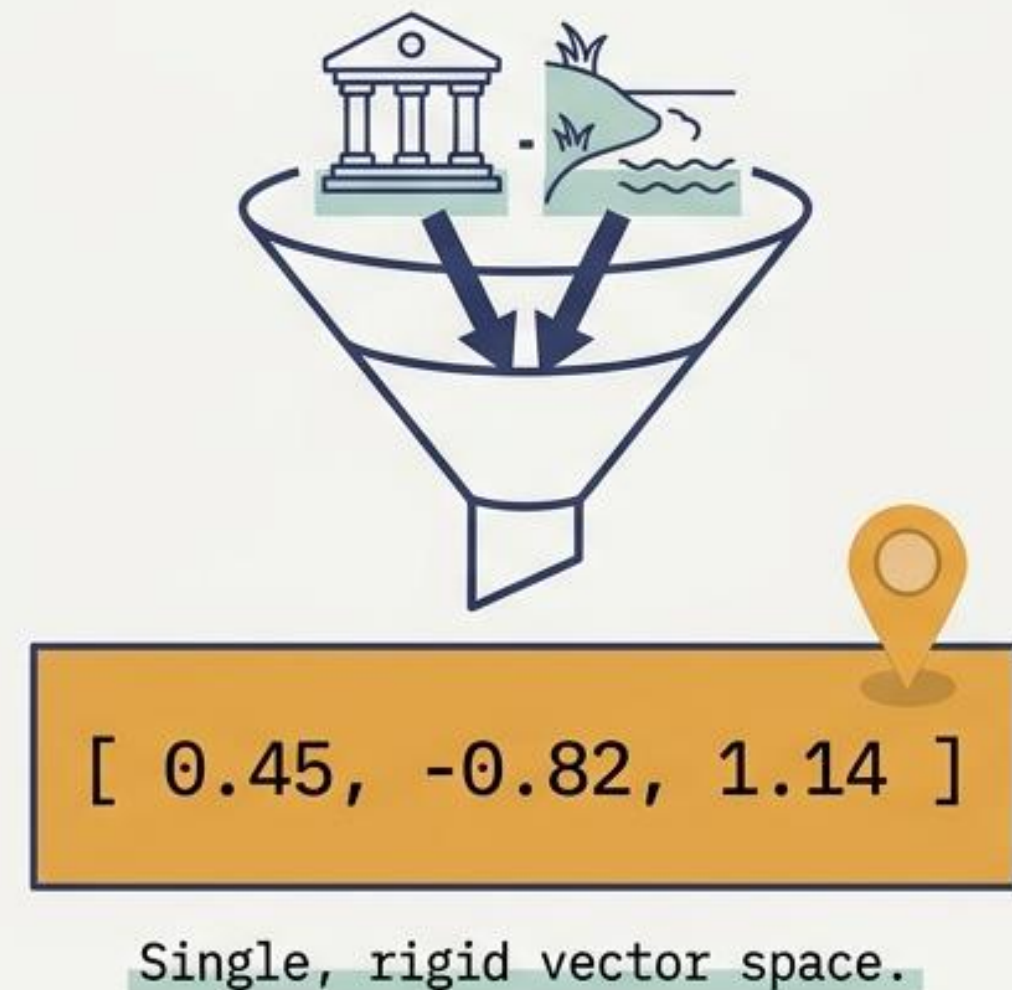


The Polysemy Problem: When Words Are Chameleons

The Ambiguity



The Static Failure



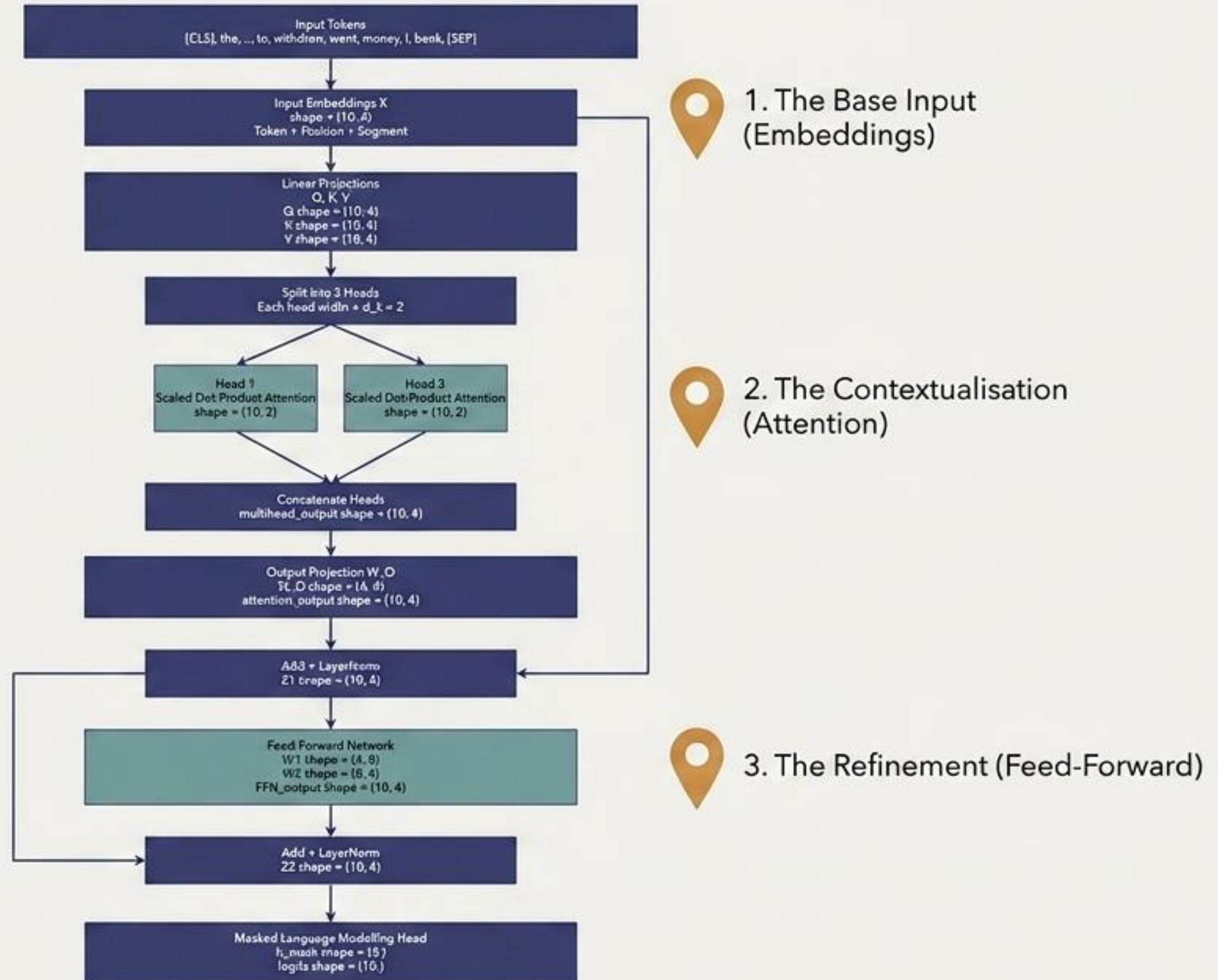
Before BERT, models struggled to separate the financial bank from the ecological bank in vector space. Context was attached after processing, not during.

The Architectural Shift: Sequential to Parallel

Criteria	LSTMs / Sequential Models	BERT / Transformers
Processing Method	Word by word (Bottlenecked)	All words simultaneously (Parallel)
Context Direction	Unidirectional (or shallow bidirectional)	Deeply Bidirectional (looks left and right instantly)
Long-Range Dependencies	Limited by sequential memory decay	Infinite range via self-attention
Core Mechanism	Recurrent Neural Networks	Multi-Headed Self-Attention

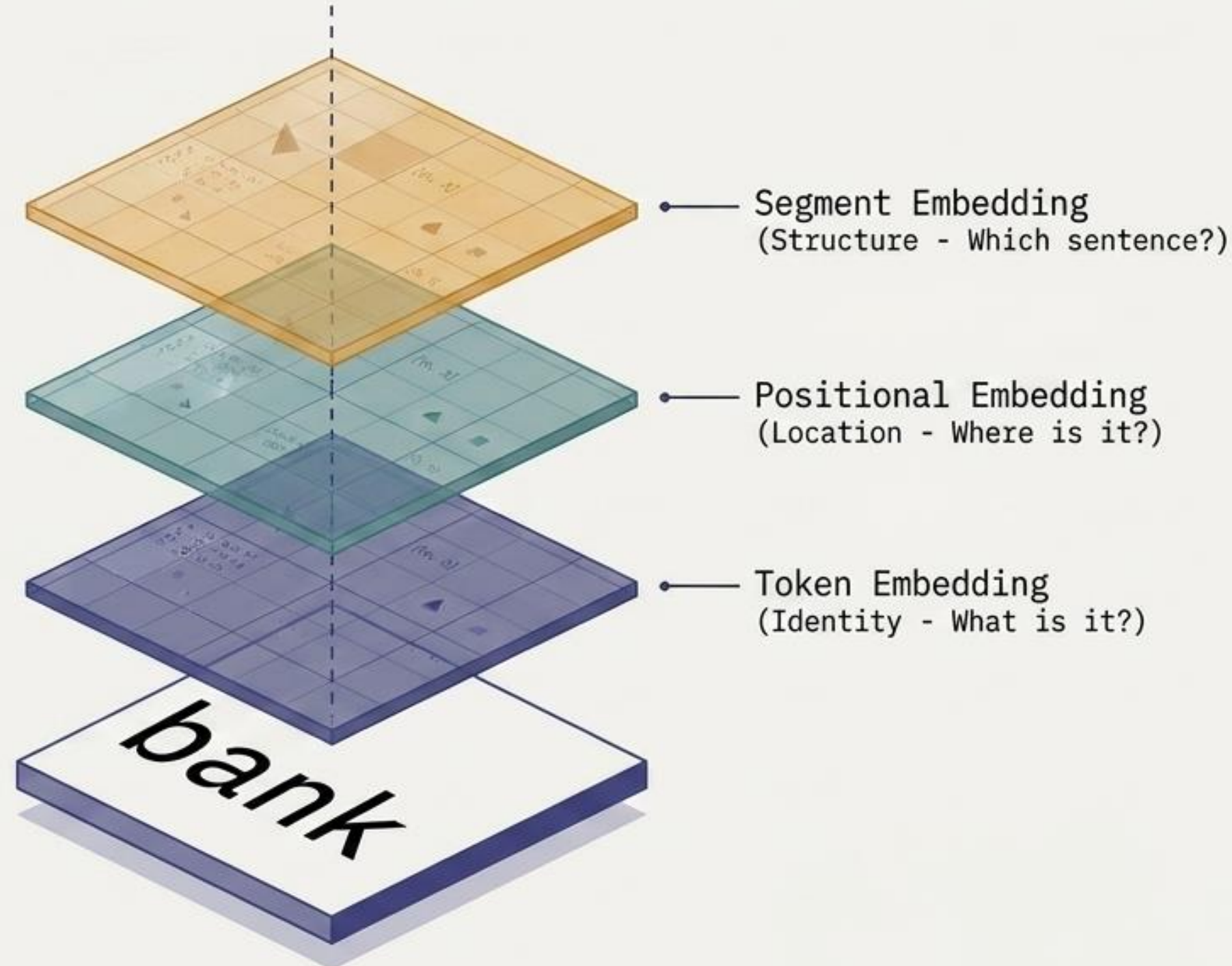
BERT fully replaces sequential recurrent networks, allowing every token to interact directly with every other token.

The Forward Pass: Anatomy of an Idea



Phase 1: Constructing the Base Embedding

$$X = \text{token_embeddings} + \text{position_embeddings} + \text{segment_embeddings}$$



Because Transformers process all words at once, order is not inherently understood.

Position must be explicitly injected as mathematical data.

Phase 2: The Attention Metaphor (Q, K, V)

The input embedding X is projected into three distinct vector spaces using learned weight matrices.

$$W_Q, W_K, W_V$$

Query (Q)



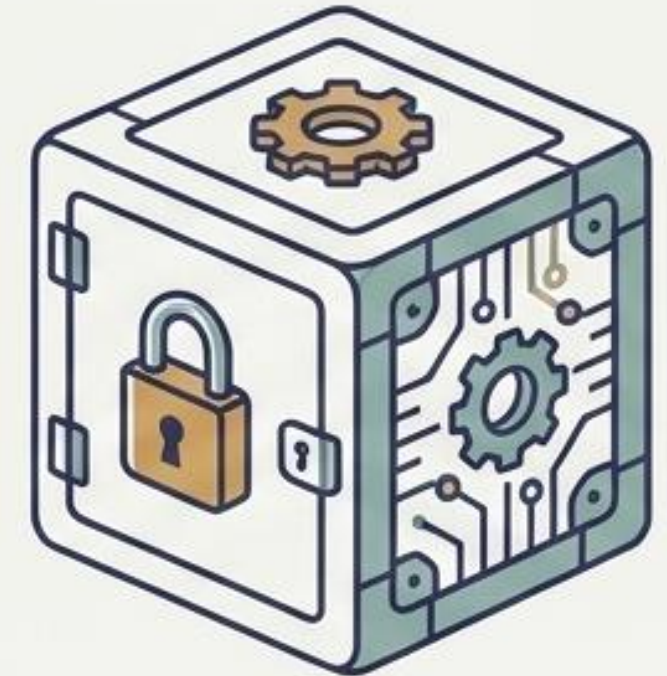
What a word is looking for in the rest of the sentence.

Key (K)



What that word offers to other words.

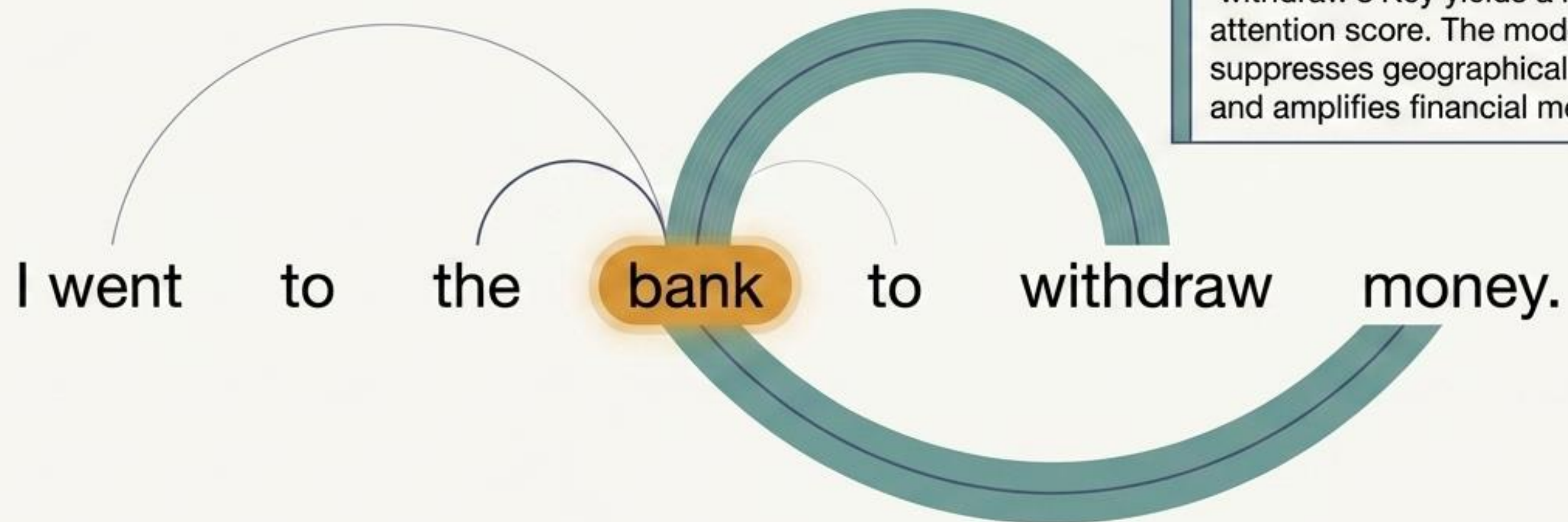
Value (V)



The actual information content passed along if attention is assigned.

Math to Meaning: Scaled Dot-Product Attention

$$\text{softmax} \left(\frac{Q * K^T}{\sqrt{d_k}} \right) * V$$



The dot product of 'bank's Query with 'withdraw's Key yields a high attention score. The model suppresses geographical meaning and amplifies financial meaning.

Multi-Headed Attention: Parallel Lenses

Head 1: Semantics

I went to the **bank** to **withdraw money**.



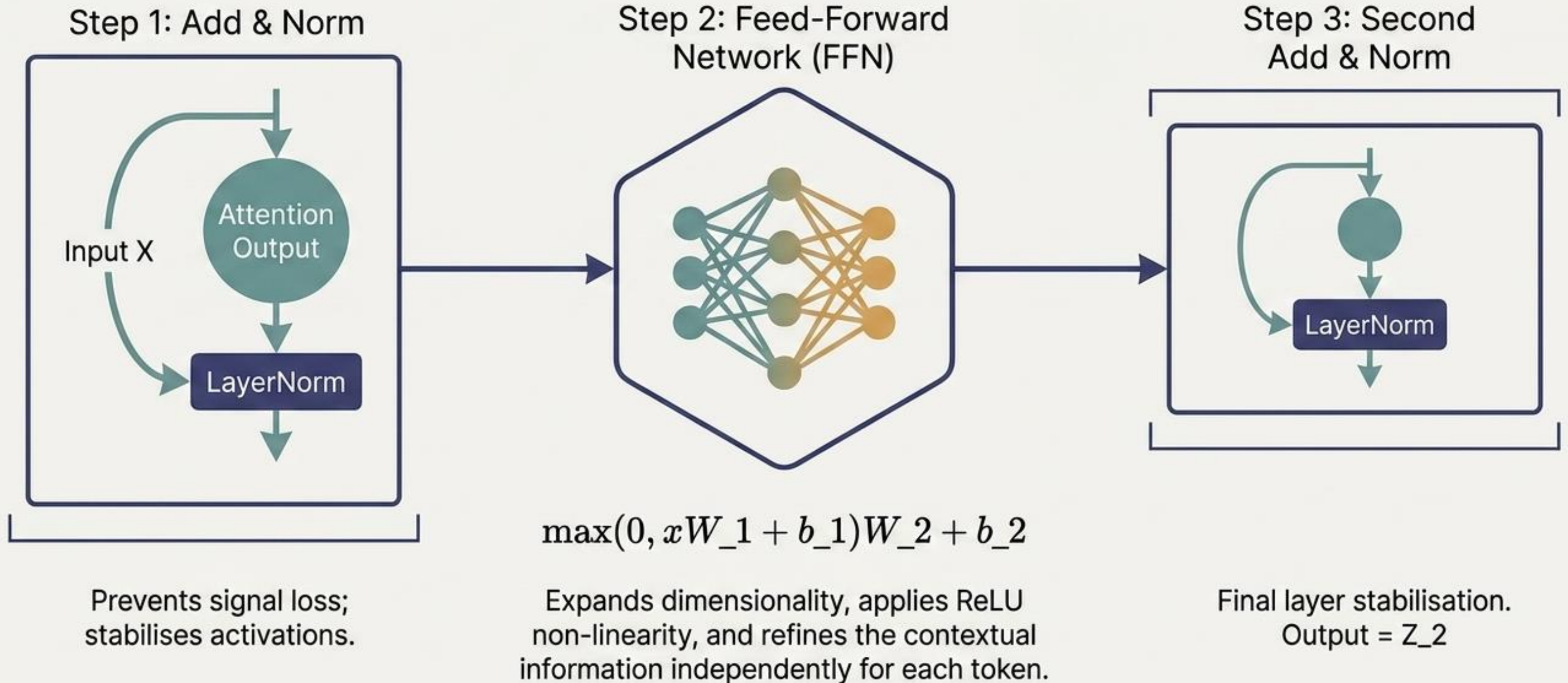
Head 2: Grammar

I **went to** the **bank** to withdraw money.



BERT computes attention multiple times in parallel. One head may master grammar, another semantic association, another positional distance. The outputs are concatenated, creating a devastatingly expressive representation.

Phase 3: Stabilisation and Transformation



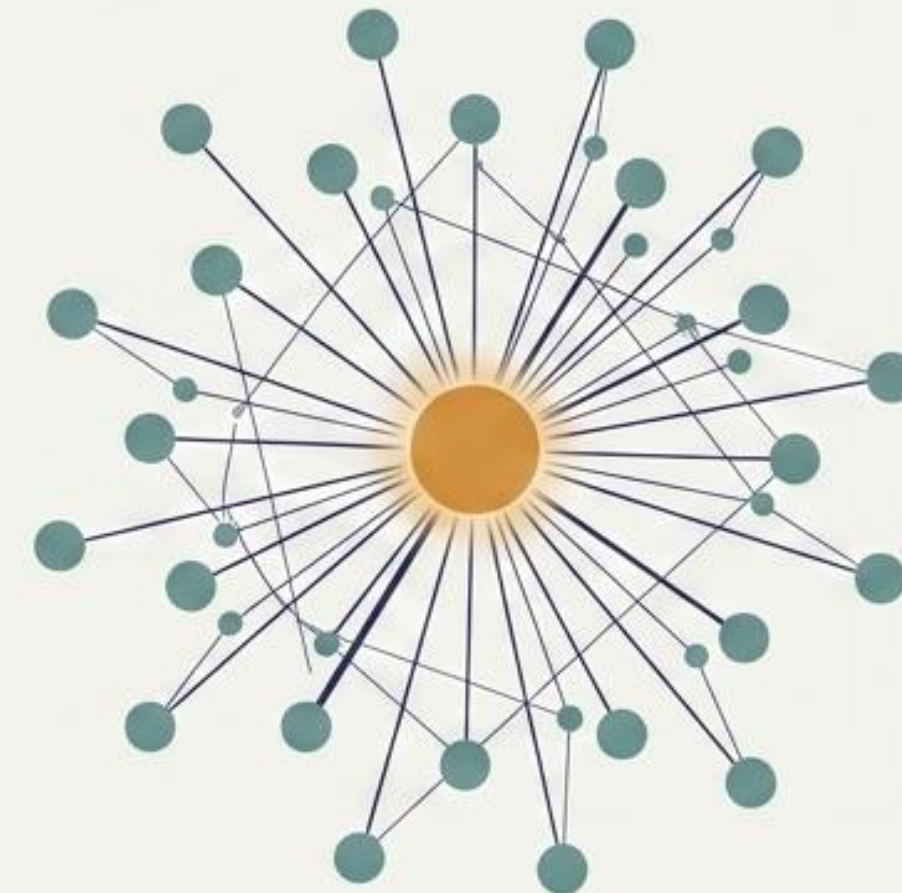
The Output: A Networked Token

Before



“A word is no longer represented in isolation—it is represented in relation to the entire sentence. Meaning is negotiated.”

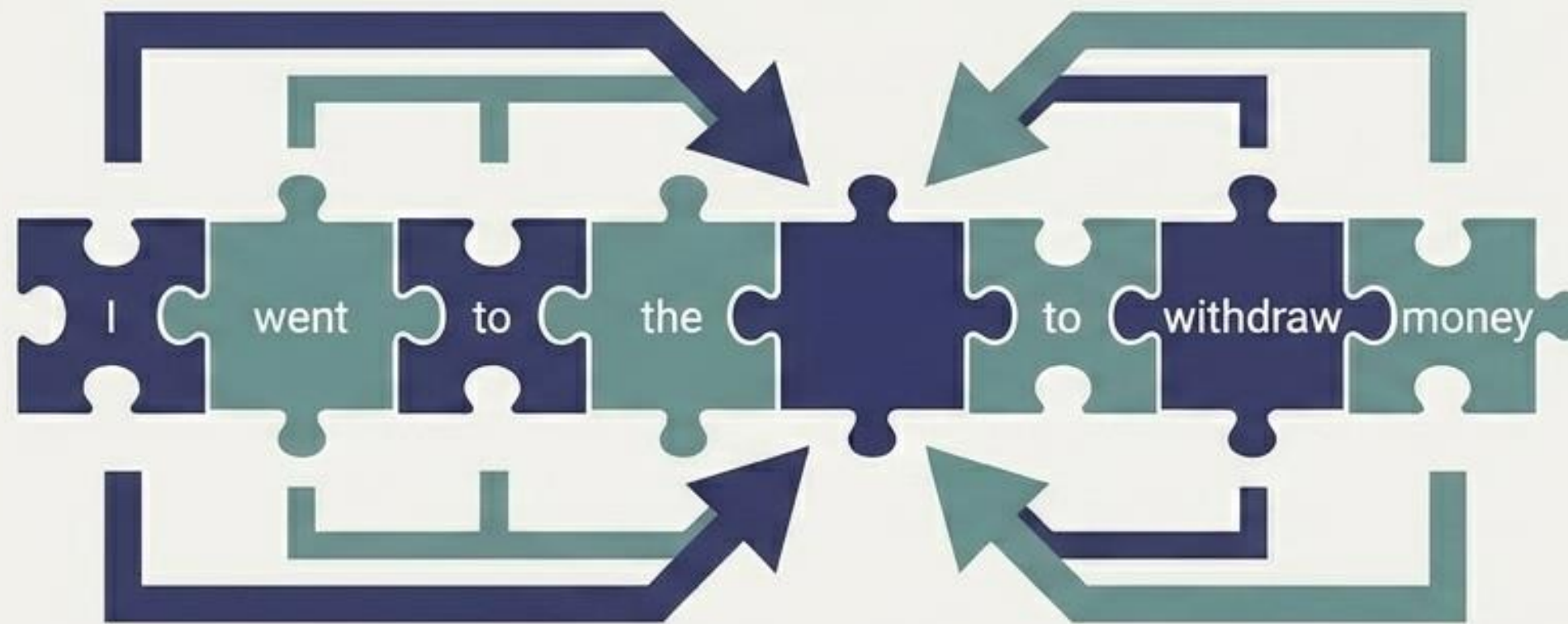
After



Input X: Static dictionary definition.

Output Z₂: Dynamically negotiated meaning.

From Computation to Understanding: The Training Paradigm



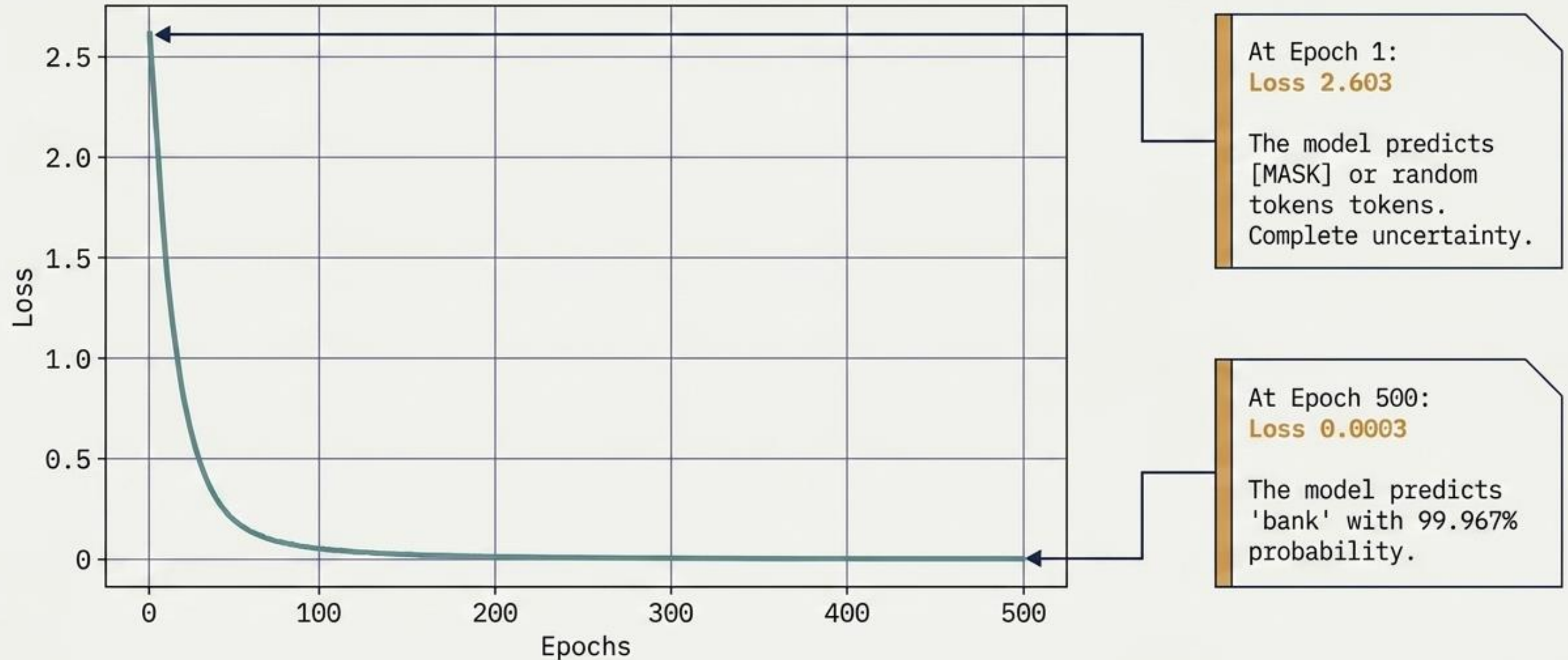
Masked Language Modelling (MLM)

15% of input tokens are randomly masked during pre-training.

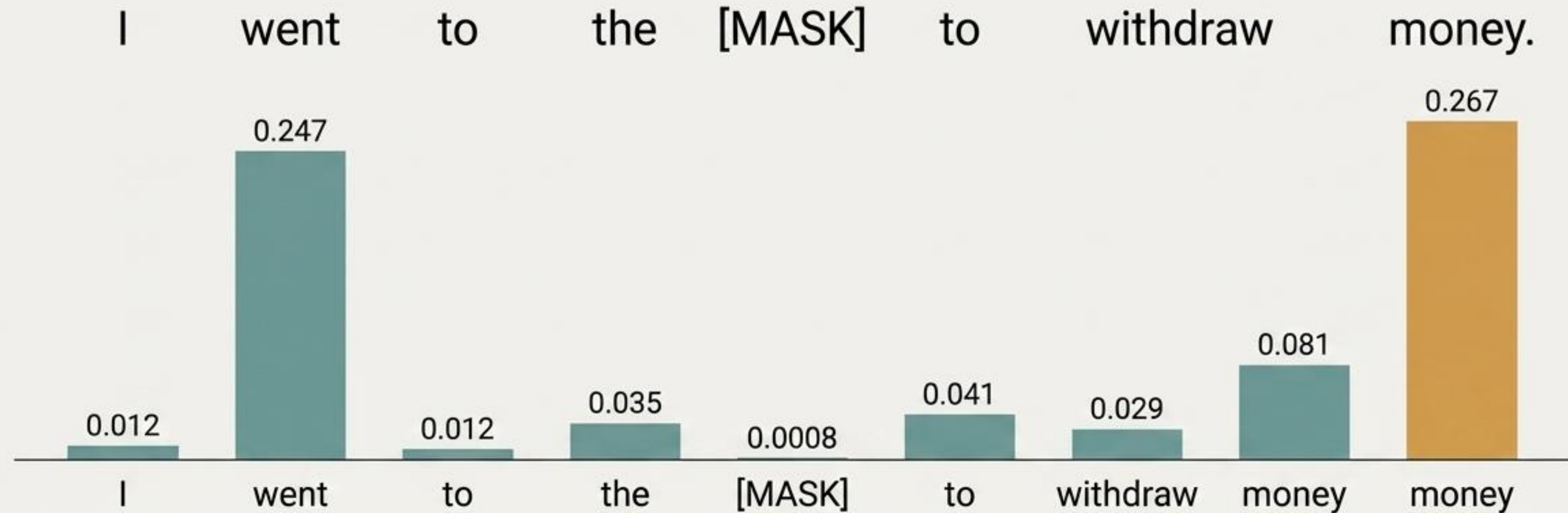
The model must predict the hidden word based purely on its left and right context.

Crucial Detail: Because self-attention is bidirectional, the model looks both forward and backward without 'cheating'.

The Proof of Learning: Loss Curve

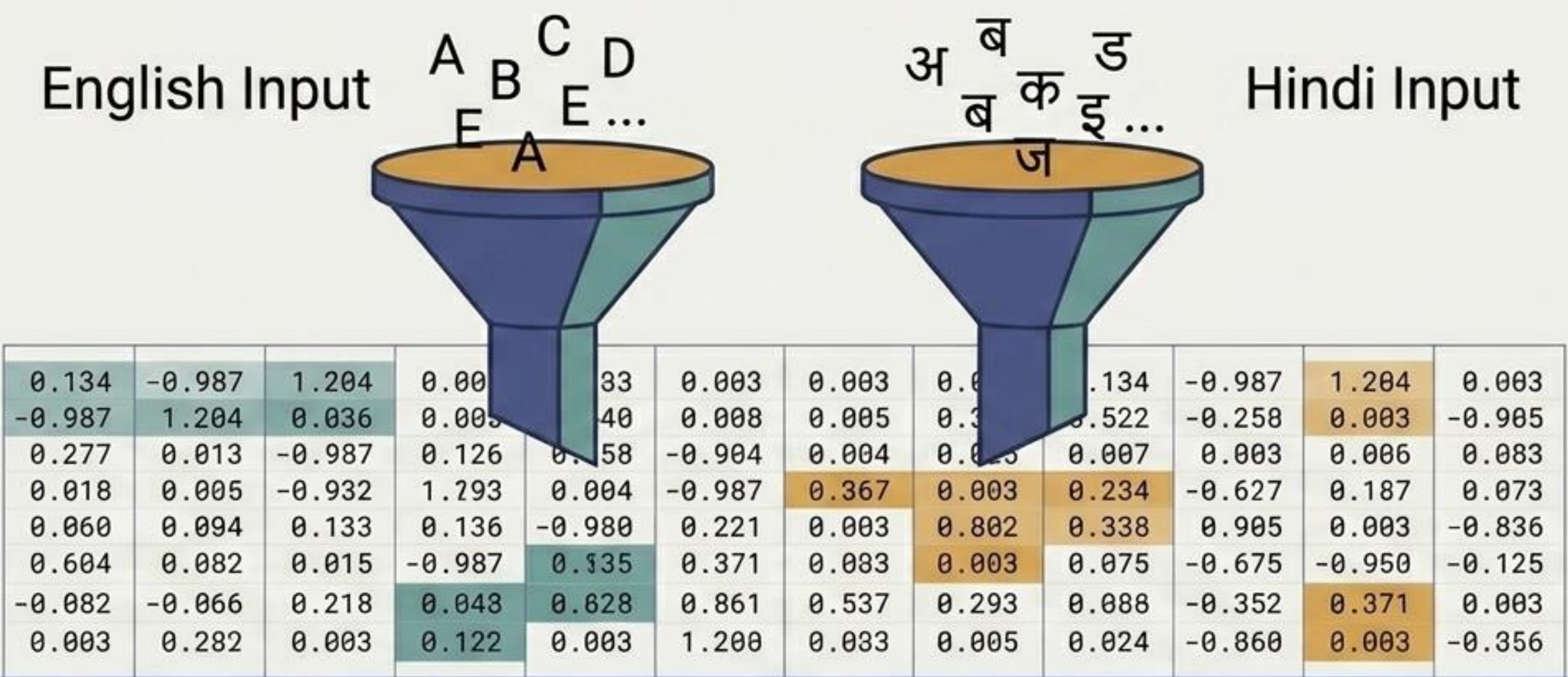


Inside the Black Box: Attention After Training



The model does not memorize. It learns to heavily weight semantic clues ('money') and positional/action clues ('went') to triangulate the missing concept.

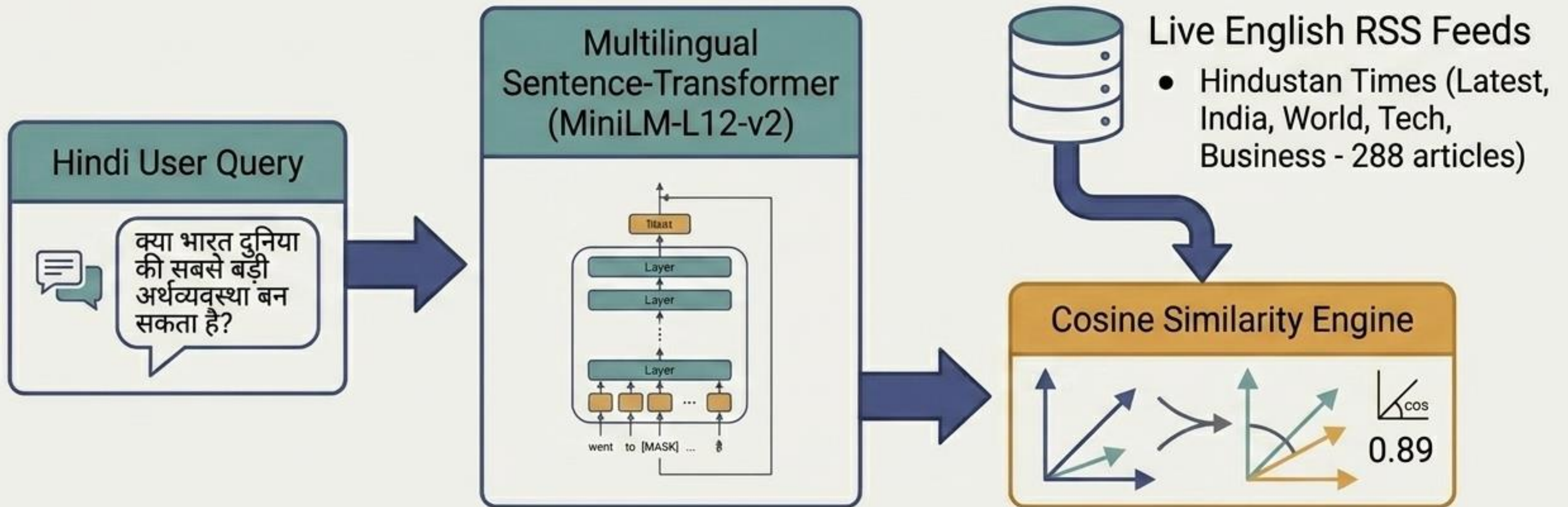
A Universal Space of Meaning



Extensive research reveals that BERT aggressively partitions words into distinct semantic clusters. Once meaning is captured in this shared embedding space, language itself becomes secondary.

How do we use this for real-time, cross-lingual semantic search?

Application: Live Semantic Search



The system embeds both the Hindi query and the English news articles into the same dimensional space and retrieves nearest neighbours.

Cross-Lingual Retrieval in Action

🔍 User Query: कृत्रिम बुद्धिमत्ता और नई तकनीक (Artificial Intelligence and New Technology)

Source: HT Technology

No to laissez-faire on AI, yes to a light touch

Match Score: 0.4789

Source: HT Technology

Samsung Galaxy A57 review: premium mid-ranger with AI smarts

The model successfully bridges the linguistic gap. It retrieves English articles about AI regulation and robotics based on conceptual meaning, not exact word matching.

Mapping Nuance: Politics and Conflict

High Alignment

Query

🔍 भारत की राजनीति (Indian Politics)

Top Result

In Karnataka, the quiet demise of Ahinda politics (Score: 0.677)

Insight

- High scores indicate the embedding space has effectively clustered political discourse tightly together.

Semantic Drift

Query

शेयर बाजार और अर्थव्यवस्था (Stock Market & Economy)

Results

Result 1: Sensex, Nifty face tug-of-war...

Result 2: ...analysts split on AJ Brown trade (Sports trade)

Insight

- Semantic search is probabilistic. A sports 'trade' shares contextual vectors with financial markets, showing the statistical nature of the model's intelligence.

The Synthesis: Statistical Intelligence

Meaning is Negotiated: Language is processed as a structure to be attended to, not a sequence to be traversed.

Unified Space: BERT dissolves boundaries between human languages, creating a purely mathematical space of semantics.

The Boundary of Encoders: BERT excels at representation, classification, and retrieval. It reads, compares, and contextualises.

If BERT represents the mastery of understanding, what happens when a model is no longer confined to interpreting text, but can create it? The transition to Generation is the next frontier.

Support Independent Research: Fueling Discovery

The Financial Barriers to Research



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